

25 July 2025

Marine Advisory: 17/2025

SUBJECT: Guidelines for test-bed and onboard measurements of methane (CH₄) and/or nitrous oxide (N₂O) emissions from marine diesel engines

Ref:

- (a) **Maritime Regulation 2.35**
- (b) **[Amended Chapter 5 – Regulations on the IMO Net-Zero Framework of the Draft 2025 revised MARPOL Annex VI](#)**
- (c) **[MEPC Resolution 391\(81\)](#)**
- (d) **[MEPC Resolution 402\(83\)](#)**

Dear Shipowners/Operators/Recognized Organizations:

Purpose

The purpose of this Advisory is to inform all Shipowners, Operators, and Recognized Organizations (RO's) of the recent guidelines that IMO has issued for measuring methane (CH₄) and/or nitrous oxide (N₂O) emissions from marine diesel engines.

Background

[Resolution MEPC 402\(83\)](#) adopted guidelines for measuring CH₄ and/or N₂O emissions from marine diesel engines in support of [MEPC Resolution 391\(81\)](#), the 2024 Guidelines on Life Cycle GHG Intensity of Marine Fuels (2024 LCA Guidelines).

The 2024 LCA Guidelines provide, among other things, guidance on life cycle GHG intensity assessment for all fuels used on board a ship and aim to cover the entire fuel life cycle (with specific boundaries), from feedstock extraction/cultivation/recovery, feedstock conversion to a fuel product, transportation as well as distribution/bunkering, and fuel utilization on board a ship.

Implementation

The Administration authorizes its RO's, when requested by an engine manufacturer, shipowner or operator, to carry out the protocol for test-bed and onboard measurements, calculation and reporting of CH₄ and/or N₂O emission values from marine diesel engines, as well as issue the documentation and verification of CH₄ and/or N₂O emission values, including the issuance of the "FORM OF STATEMENT OF EMISSION VALUES FOR METHANE (CH₄) AND/OR NITROUS OXIDE (N₂O)" in the form provided in Appendix 3 of [Resolution MEPC 402\(83\)](#).

For more information, please contact the Regulations and Standards Department at RegsandStandards@liscr.com or telephone +1 703 790 3434.

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